

Objects

Introduction to Programming

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Introduction

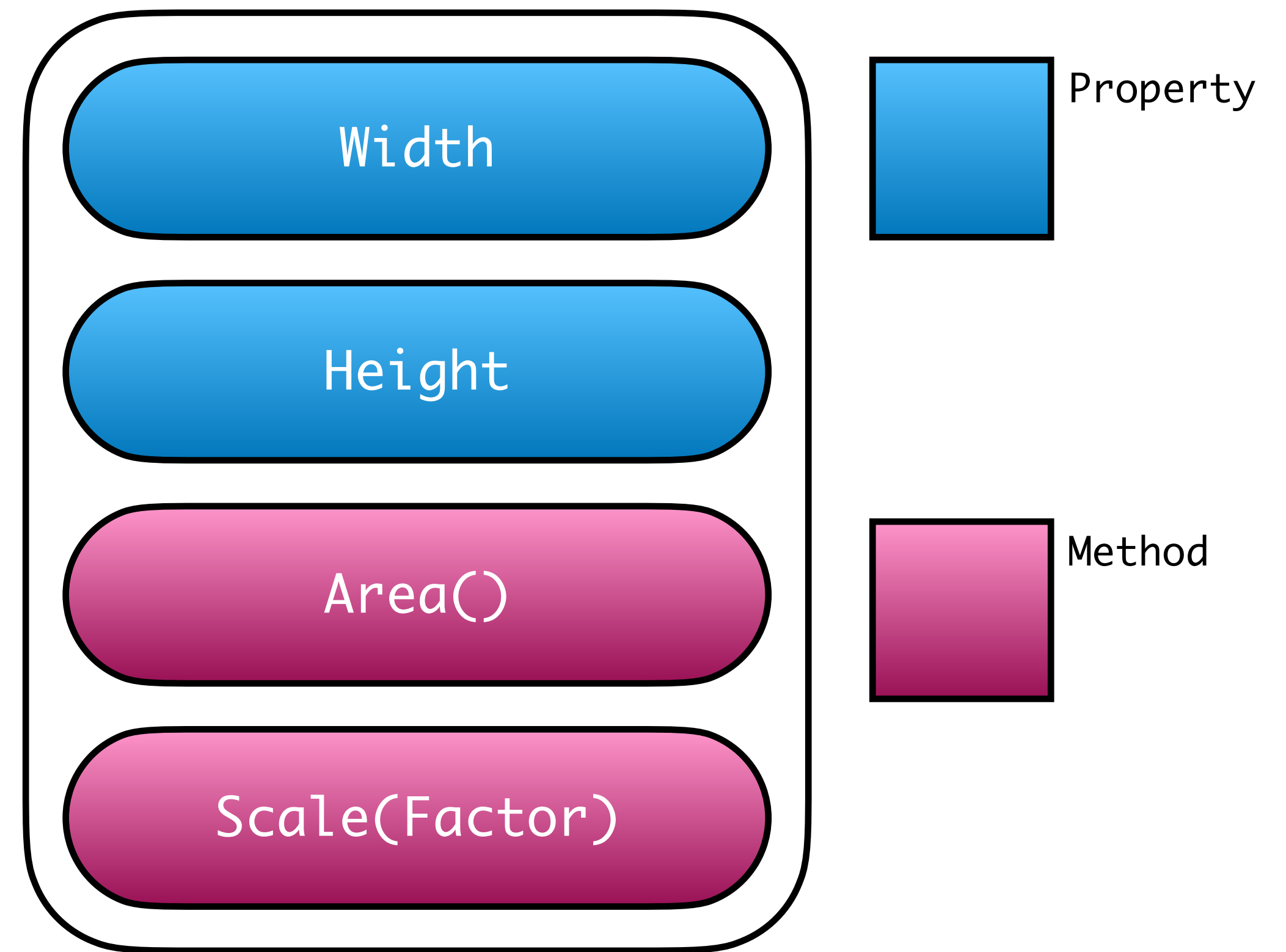
Objects

- **Object:** a group of related variables and functions
- Variables in an object are called attributes/fields/properties
- Functions in an object are called methods
- An object's methods should use/change its properties

Introduction

Objects

- A rectangle has a width & height
 - These are its properties
 - An object's props and their values represent the object's state
- A rectangle's area is computed using its props (so area must be a method)
- The scale method updates the object's props



Classes

Objects

- **Class:** a template for creating objects; a data type
- The class dictates what properties & methods each object of its type has
- The class does NOT set the values of its properties (each object is different)

Classes

Objects

- PY: `class Rectangle:`
- PY: `def __init__(self, width, height):`
- PY: `self.width = width`
- PY: `self.height = height`

Classes

Objects

- JS: `class Rectangle {`
- JS: `constructor(width, height) {`
- JS: `this.width = width;`
- JS: `this.height = height;`
- JS: `}`
- JS: `}`

Classes

Objects

- JV: `class Rectangle {`
- JV: `int width;`
- JV: `int height;`
- JV: `Rectangle(int width, int height) {`
- JV: `this.width = width;`
- JV: `this.height = height;`
- JV: `}`
- JV: `}`

Classes

Objects

- C#: `class Rectangle {`
- C#: `int Width;`
- C#: `int Height;`
- C#: `Rectangle(int width, int height) {`
- C#: `Width = width;`
- C#: `Height = height;`
- C#: `}`
- C#: `}`

Object Creation

Objects

- PY: `rect1 = Rectangle(6, 4)`
- -----
- JS: `var rect1 = new Rectangle(6, 4);`
- -----
- JV: `Rectangle rect1 = new Rectangle(6, 4);`
- -----
- C#: `Rectangle rect1 = new Rectangle(6, 4);`

Getting Properties

Objects

- PY: `print(rect1.width)`
- _____
- JS: `console.log(rect1.width);`
- _____
- JV: `System.out.println(rect1.width);`
- _____
- C#: `Console.WriteLine(rect1.Width);`

Setting Properties

Objects

- PY: `rect1.width = 8`
- _____
- JS: `rect1.width = 8;`
- _____
- JV: `rect1.width = 8;`
- _____
- C#: `rect1.Width = 8;`

Declaring Methods

Objects

- PY: `class` Rectangle:
- PY: `...`
- PY: `def` area(self):
- PY: `return` self.width * self.height

Declaring Methods

Objects

- JS: `class Rectangle {`
- JS: `...`
- JS: `area() {`
- JS: `return this.width * this.height;`
- JS: `}`
- JS: `}`

Declaring Methods

Objects

- JV: `class Rectangle {`
- JV: `...`
- JV: `int area() {`
- JV: `return this.width * this.height;`
- JV: `}`
- JV: `}`

Declaring Methods

Objects

- C#: `class Rectangle {`
- C#: `...`
- C#: `int Area() {`
- C#: `return Width * Height;`
- C#: `}`
- C#: `}`

Calling Methods

Objects

- PY: `print(rect1.area())`
- _____
- JS: `console.log(rect1.area());`
- _____
- JV: `System.out.println(rect1.area());`
- _____
- C#: `Console.WriteLine(rect1.Area());`

Object Oriented Programming

Objects

- An approach to program design & development focused on things
- The program's functionality is grouped into objects alongside the data said functions operate upon
- Problem description nouns might indicate classes